

Gaétan Lepage

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*PhD in deep learning for robotics. Nix/NixOS contributor.
Looking for a job in robotics, HPC or development tooling.*

Education

 **Grenoble Alpes University**
PHD IN DEEP LEARNING FOR ROBOTICS

Grenoble, France
2020-2025

- **Title:** From Sound to Action: Deep Learning for Audio-Based Localization and Navigation in Robotics
- **Supervisors:** Dr. Xavier Alameda-Pineda, Pr. Laurent Girin and, Dr. Chris Reinke
- Defended in July 2025

 **Grenoble Alpes University**
MASTER OF SCIENCE IN INDUSTRIAL AND APPLIED MATHEMATICS

Grenoble, France
2019 - 2020

- Data Science track: machine learning, optimization, optimal transport, probabilities, computer vision
- Grade: 16.32/20 | Rank: 2/50

 **Ensimag, Grenoble Institute of Technology**
ENGINEERING SCHOOL IN COMPUTER SCIENCE AND APPLIED MATHEMATICS

Grenoble, France
2017 - 2020

- MMIS track (Mathematics Modeling, Image & Simulation)
- Courses:
 - **Applied Mathematics:** functional analysis, probabilities, optimization, statistics, dynamical systems
 - **Computer Science:** software engineering, language theory, algorithms and data structure

Internship and work experience

 **Nix CUDA team**
TEAM MEMBER

Remote
2025 - now

- Daily maintenance tasks on CUDA-related packages
- Responsible for the recent [NixOS-CUDA infrastructure](#) (Hydra CI, binary cache, CPU builders, GPU hosts for testing)
- **Team presentation:** <https://nixos-cuda.org/>

 **Ensimag, Grenoble Institute of Technology**
TEACHING ASSISTANT

Grenoble, France
2020 - 2022

- Teaching Assistant at Ensimag for first-year engineering students
- 4 courses (120h total):
 - Statistical principles and methods
 - Applied probabilities
 - Software development project in C (implementing a GUI library in C)
 - Basics of imperative programming (in Python)

 **University of Applied Sciences and Arts of Western Switzerland**
MASTER'S THESIS, RESEARCH INTERNSHIP

Sierre, Valais, Switzerland
2020 - 2020

- Five-month research internship
- **Thesis:** Segmentation of compound figures in medical publications using deep learning

 **Ensimag, IT department**
BUGBUSTER

Grenoble, France
2018 - 2020

- Paid student role providing IT support to Ensimag students
- Installation and maintenance of Linux environments (dual-boots, virtual machines, troubleshooting, etc.)

Open source projects

nixpkgs

SOFTWARE REPOSITORY FOR THE Nix(OS) ECOSYSTEM

- One of the largest GitHub repos (~750k commits, ~2 million Nix LoC) and package repositories (100k packages)
- I maintain 200 packages, including PyTorch, protobuf, Jax, and Dask
- I interact with multiple build toolchains and languages: C/C++ (Cmake, make and meson), Rust (cargo), Python (setuptools, hatchling), Go etc
- I have authored more than 2,800 PRs and reviewed more than 3,000
- **GitHub repository:** <https://github.com/NixOS/nixpkgs>

Nix

Acoustix

SIMULATION LIBRARY FOR ROBOTICS AUDITION

- Dynamic acoustic simulation for robotic navigation and localization
- Developed during my PhD @Inria
- **GitHub repository:** <https://github.com/GaetanLepage/acoustix>

Python

RemI (Remote Inria)

CLI TOOL FOR INTERACTING WITH A COMPUTE CLUSTER

- Allows scientists from the RobotLearn team to easily interact with Inria's computing infrastructures:
 - Seamlessly sync project files to a remote storage partition
 - Run jobs on remote compute nodes or workstations (using the OAR scheduler)
 - Build and interact with Singularity/Apptainer containers
- **GitLab repository:** <https://gitlab.inria.fr/remote-inria/remi>
- **Documentation:** <https://remote-inria.gitlabpages.inria.fr>

Python

Cluster monitor

WEB-BASED LIVE VISUALIZATION FOR THE CLUSTER INFRASTRUCTURE

- Provides an effective and concise overview of a GPU cluster's occupancy.
- **GitLab repository:** <https://gitlab.inria.fr/robotlearn/cluster-monitor>

Python

Nixvim

COLLECTION OF NIX MODULES AND TOOLING FOR CONFIGURING NEOVIM USING NIX

- 2,000 stars on GitHub, 100k lines of code (tests included)
- I have been a core maintainer for the past 3 years
- **GitHub repository:** <https://github.com/nix-community/nixvim>

Nix

Skills

Languages	English, French
Programming languages	Python, Nix, Bash, C (basics), Java (basics)
Tech Stack	• Deep Learning stack (PyTorch, NumPy) • Containerization (Apptainer/Singularity, Docker) • Nix/NixOS • Software building and packaging (make, meson, CMake, cargo) • Linux system administration • Git, UNIX tooling